

Digital Finance

Winter 2025/2026

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DESCRIPTION

Blockchain has proven its product-market fit as a foundational technology over the last few years. It will become a ubiquitous technology across consumer and enterprise use-cases within the next few years. The financial sector is the fastest to adopt that revolutionary stack, be it in web3-native or traditional finance businesses. Concurrently, GenAI has taken the consumer space by storm, including for professional use-cases. Yet, the challenges to apply it to financial services at scale remain steep, and expectations remain high.

The ecosystem needs talent to go invent, build, and commercialize the future of financial services. This class provides a core knowledge and understanding to go do that.

OBJECTIVES

At the close of this course, you will have acquired a good understanding of core pillars of blockchain and AI in finance, and a critical thinking on use-cases towards separating fads from foundational projects as well as a critical thinking on key considerations for product and GTM.

These will make you competent to work on digital finance use-cases and opportunities where a solid understanding of blockchain and GenAI will be needed to succeed. Roles you will be able to work in will cover both business or product capacity, at a financial institution, fintech, or blockchain project.

MATERIALS

Recommended

- Curve whitepapers <https://www.curve.fi/stableswap-paper.pdf>
- JPM Onyx materials <https://www.jpmorgan.com/onyx/index>
- Uniswap v3 whitepaper <https://uniswap.org/whitepaper-v3.pdf>
- Basel Committee Framework on the Regulation of Digital Assets <https://www.bis.org/bcbs/publ/d545.htm>
- Rekt articles <https://rekt.news>
- Commissioner Pierce statements <https://twitter.com/HesterPeirce>
- Blockchain Association Policy resources <https://theblockchainassociation.org/policy/>
- Lido whitepaper <https://www.securities.io/lido-dao-whitepaper/>
- Court appointee's reports on the bankruptcies of FTX, Celsius, Genesis
- Headline news on blockchain from serious media (Bloomberg, CoinDesk, Forbes, Rekt)

SCHEDULE

Every Friday from 9am PST till 11:50am PST, starting second week of January 2026, for the whole trimester. Finals are open-resources, a mix of multiple-choice questions and open questions, individually graded.

One class will need to be skipped mid-March given a conflicting event. You can use that time slot to review all your notes and the key slides.

Date	Class Topic & Activities	Assignments
Week 1	Fundamentals of Financial Services: Major Workflows	
Week 2	Fundamentals of Financial Services: Payment Rails	
Week 3	Fundamentals of Financial Services: Venture Capital Case Study: How Can We Reinvent Banking?	Make sure to have reviewed all the previous slides and come prepared to discuss during the case study
Week 4	Fundamentals of Blockchain Technologies: Regulatory Frameworks, Core Technology Elements Fundamentals of AI Technologies: Regulatory Frameworks, Core Technology Elements	
Week 5	Overview of Decentralized Finance: Infrastructure & Important Use Cases	
Week 6	Stablecoins	
Week 7	NFT and Real-World Assets	
Week 8	Fintech 3.0: Important Use Cases in Neobanking, Cross-Border, Payment, Lending, Asset Management	
Week 9	Fintech 3.0: Important Use Cases in Supply-Chain Career Tracks	
Week 10	No class	Review all the key slides + your notes on them before the final
Week 11	Open-resources Final	

ASSIGNMENTS

Reading Assignments

Read material when asked to, and come to class prepared to discuss and engage

Team Projects

During the third class discussing venture capital, we will run a real-time case study diving into how to reinvent banking

Quizzes

None

Exams

Individual exam, with a mix of multiple-choice questions and open questions

Class Participation

- Is essential and will go towards the final grade
- If a student misses at least 2 classes by not being present, and did not notify ahead of time of those absences, the student will be dismissed from the class effective immediately.
- Participation doesn't consist in just showing up to class, it consists in asking questions and sharing opinions towards attaining a stronger understanding of the concepts introduced in class

GRADING

Assignments	Points [or percentage]
Individual final exam	50%
Real-time Case Study	25%
Class participation	25%
Total	100

Students can choose to audit the class, and it is Letter graded otherwise.

COURSE POLICIES

Students are expected to attend, and to participate in a constructive manner.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.